

Team Number: _____ School & Team Name: _____

Student Names: _____

IMPOUND

1. T F 5.a. **IMPOUND (At State and National Tournaments only, NOT at regionals):** The team impounds their device and any tools or parts that they will use during their set-up time before the first time slot. **If false, Tier 3.**
2. T F 6.c. The ASL is submitted on time at Impound. (Or at Regionals, submitted during check-in)

CHECK-IN AND SAFETY

3. T F 2.a. Eye Protection: Participants properly wear Class C eye protection at all times. **If F, Tier 3 and fill out Box 5, 6, and 7.**
4. T F 2.d,5.c.iii Device passes an ES safety inspection before operation. Device has no potential hazards or safety concerns, including uncontrollable flames, cell phones, hazardous liquids, gases and materials (e.g., rat traps, lead objects, combustible fuses, dry ice, liquid nitrogen, etc.) All chemicals, if used, are safely handled. **Participants are not permitted to run until the ES is satisfied with safety compliance.**

Construction Parameters:

5. _____ 3.a, 6.l **Device Dimension 1:** Dimension 1 of device to the nearest 0.1 cm (80 cm dimension)(Penalty for over 80cm and >82 = Tier 2)
6. _____ 3.a, 6.l **Device Dimension 2:** Dimension 2 of device to the nearest 0.1 cm (80 cm dimension)(Penalty for over 80cm and >82 = Tier 2)
7. _____ 3.a, 6.l **Device Dimension 3:** Dimension 3 of device to the nearest 0.1 cm (80 cm dimension)(Penalty for over 80cm and >82 = Tier 2)
8. T F 3.b. The top and at least two vertical walls are open or transparent for viewing all actions.
- T F 3.c. Parallel and dead-end actions are not used.
- T F 3.d. Each moveable/adjustable physical object in the device must be utilized by at most one assigned action. An object at the end of one action may initiate the next action but is not used in any further action.
- T F 3.g. No Arduinos, Raspberry Pis, or integrated circuits are used, and timing is not powered by electricity or springs.
- T F 3.h. Only commercial batteries, not exceeding 9 volts as labeled, are used to energize each of the Device's electrical circuits. The expected voltage output across any two points does not exceed 9 volts as calculated using their labeled voltage. All energy storage devices must be contained in the device, and comply with the Science Olympiad Battery Policy at www.soinc.org.

9. T F ALL CONSTRUCTION PARAMETERS ARE MET

10. T F 6.n. **Remote Control:** Device is not remotely timed or controlled. **If F, team gets participation points only.**

Competition Parameters:

11. T F 5.c.v. Team is able to answer questions on design, construction, and operation of the device per the Building Policy found on www.soinc.org. **If F, Tier 3.**
12. T F 5.c.i. Once participants start the Check-In process, they do not leave or gain any outside assistance or materials until the Event Time is completed. **If F, Tier 2.**

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SET-UP AND ASL:

13. T	F	6.b.	Participants use $\leq$ 30 minutes to set-up their device.
14. T	F	6.b.i.	State and Nationals Only: Participants use $\leq$ 15 minutes to set-up their device.
15. T	F	6.d.	The ASLs presented are in proper format.
16. T	F	6.e.	All scoreable and non-scoreable actions are included and accurately described.
17. T	F	6.f.	The original number (column 1) in the ASL is properly labelled in the Device and the action number (column 3) is the roman numeral of the 12 optional intermediate tasks (I to xii).

DEVICE OPERATION:

18.

1	2	4.a.	Start Action: Circle "1, 2, 3, 4, or Touch" based on how many coins are used to initiate the first action. Requirement: Device has a vertical coin slot on one side that is only large enough to allow one coin to be added at a time. - If the team must touch the device to start it, no points are awarded for the start and a penalty for the first touch is assessed.
3	4		
TOUCH			

Scoreable Actions:

T	F	4.b.i.	Vinegar and baking soda is used to inflate a balloon so that it expands 20 cm, then the balloon strikes an object that originally was at least 20 cm above the balloon, causing the object to initiate the next action.
T	F	4.b.ii.	An endothermic action initiates the next action because of the reduction in temperature.
T	F	4.b.iii.	Two Alka Seltzer tablets are dropped into water so that the reaction triggers the next action.
T	F	4.b.iv.	A 1st class lever is used to launch an unaltered ping pong ball completely out of the top of the Device, flies at least 20 cm horizontally before re-entering, initiating the next action, and staying in the Device. A guide of 20 cm is included at the top level of the Device.
T	F	4.b.v.	An infrared beam is used where the transmitter and receiver are at least 20 cm apart to initiate the next action. The transmitter is an IR LED and the receiver is a photodiode-based receiver.
T	F	4.b.vi.	An object with a mass $\geq$ 400 g is pushed or pulled up an inclined plane with an IMA $\geq$ 2 at least 10 vertical cm before the object initiates the next action. The mass is removeable for the ES to verify.
T	F	4.b.vii.	A pulley system with an ideal mechanical advantage (IMA) of at least 3 is used to lift an object with a mass $\geq$ 400 g at least 10 vertical cm before the object initiates the next action. The mass is removeable for the ES to verify.
T	F	4.b.viii.	The mechanical advantage of all 3 classes of levers is used in any sequence to initiate the next action.
T	F	4.b.ix.	Gravity is used to clearly rotate a screw so that it operates as a screw converting rotational force into linear force. The screw completes at least two full rotations and moves an object at least 2 cm before that object initiates the next action. The screw is marked so its rotational movement is clearly visible.
T	F	4.b.x.	An Archimedes Screw is used to raise a marble at least 20 cm before the marble initiates the next action. The entire action is visible.
T	F	4.b.xi.	Electricity is used to directly or indirectly break a string or fishing line so that the breaking of the line drops an object that initiates the next action.
T	F	4.b.xii.	A magnet is removed from a surface so that a magnetic object falls due to the removal of the magnetic force. The falling object initiates the next action.

19. _____ # of Scorable Actions Successfully Completed from 4.d.i to 4.d.xii (Max = 12 Actions)

20. _____ 4.c. **Water Timer:** Time, in seconds, that the water timer runs before the Target Time. Stop time when the time reaches the Target Operation Time. The next action must be successfully initiated by the timer. For State/National tournaments, the team must demonstrate how this timer is adjusted to account for the increased length in target time.

FINAL ACTION

T	F	4.d	Final Action Platform: The device contains a platform made of flat, rigid, solid, unpadded material that is no more than 12 cm x 12 cm x 0.5 cm with no sides. The platform has no barriers within 4 cm of the edge. The top of the platform is at least 2 cm above the top of the base.
T	F	4.e	Final Action: After attempting all scoreable actions, the device dispenses a commercially purchased, unaltered 5-stick pack of chewing gum at least 50 cm vertically onto a platform that meets the conditions above. The dropped pack of gum is the only thing on the platform, is clearly visible and supported only by the platform. The team then removes the gum through an opening on the same side of the device as the coin slot.
21. T	F	4.d	Do not Touch: Team does not touch the action that leads to releasing the pack of gum to complete the Final Action.

22. T F ALL THREE FINAL ACTION PARAMETERS ARE MET

ADJUSTMENTS AND TIMING

<u>23.</u>		5.e.iii.	Adjustments: # of adjustments (up to 3) made to device. If the device stops/fails after the 3rd adjustment, all scoring stops. If an adjustment is made leading to a completed action, the action will not count for points. Adjustments include touches that were used for the Start and Final Action.	
		7.d.		
<u>24.</u>	T	F	7.f.	Electricity: Use of electricity is limited to the first 30 seconds of operation.
<u>25.</u>	T	F	7.e.	Solids and Liquids: do not leave the measured dimensions of the Device with the exception of the ping pong ball in action 4.b.iv.
<u>26.</u>	T	F	4.a.vi.	Coins are Returned: All coins are returned to the ES at the end of the operation of the Device. If F, Tier 2.
<u>27.</u>			5.e.i	Device Operation Time: Time, in seconds, of device operation. Begins when the first action starts after the coin is deposited into the device and stops when the pack of gum comes to rest on the platform or after 2x the Target Time in seconds has elapsed, whichever comes first. If device fails after three or more adjustments are made, the device time is 0 seconds and scoring stops.
<u>28.</u>	T	F		General Rule: The team is disqualified. (Notify the team and their coach as soon as possible.)